

GURU

Guru AI - SEE 2081 (2025) Model Solutions

Optional II Computer Science (Grade Increment - GI-3031)

Group 'A'

Ans 1.

- (a) Two unguided transmission media are Radio waves and Microwaves.*
- (b) Online payment refers to the electronic transfer of money over the internet for goods or services purchased.*
- (c) Two objects of MS-Access are Table and Form.*
- (d) A report is a database object used to summarize, format, and present data in an organized, printable manner.*
- (e) The 'WRITE' statement is used to write or record data items (variables) into a sequential data file, enclosing string data in double quotes and separating items with commas.*
- (f) Two data types used in C-language are `int` (integer) and `float`.*

Ans 2.

- (a) Cybercrime*
- (b) Bandwidth*

Ans 3.

- (a) ISP: Internet Service Provider*
- (b) G2G: Government to Government*

Group 'B'

Ans 4.

(a) Data Communication:

Data communication is the process of transmitting digital data (such as text, audio, video) between two or more computers or devices through a communication medium.

Two modes of data communication: Simplex Mode and Half-Duplex Mode.

(b) Antivirus Software:

Antivirus software is a utility program designed to detect, prevent, and remove malware and computer viruses from a computer system to keep it safe.

Examples: Kaspersky, McAfee.

(c) Software & Hardware Security:

Software security refers to protecting computer software and data from unauthorized access, modification, or destruction (e.g., using passwords, antivirus).

Two measures of hardware security:

- 1. Using UPS (Uninterruptible Power Supply) to prevent damage from power fluctuations.*
- 2. Keeping computer systems in dust-free and temperature-controlled environments.*

(d) Internet of Things (IoT):

IoT is a network of physical devices, vehicles, home appliances, and other items embedded with sensors and software that connect and exchange data over the internet.

Importance:

- 1. It enables automation and remote control of smart home devices.*
- 2. It improves efficiency in industries through real-time monitoring and data collection.*

(e) E-commerce:

E-commerce (Electronic Commerce) refers to the buying and selling of goods or services using the internet, and the transfer of money and data to execute these transactions.

Types: B2B (Business to Business), B2C (Business to Consumer), C2C (Consumer to Consumer), and C2B (Consumer to Business).

(f) Table & View:

A table is the primary building block of a database that stores data in rows (records) and columns (fields).

Design View is used to modify the structure (fields, data types, properties) of a table.

(g) Query:

A query is a database object used to retrieve, filter, or manipulate specific data from one or more tables based on given conditions.

Importance:

1. Allows users to view only the specific data they need from a large database.
2. Can be used to perform calculations or update multiple records simultaneously.

(h) Validation Rule & Validation Text:

Validation Rule: A property that specifies the criteria or conditions that the data entered into a field must meet to be accepted.

Validation Text: A property that defines the custom error message displayed to the user when they enter data that violates the Validation Rule.

(i) Form vs Report:

1. A Form is primarily used for data entry, editing, and viewing records on the screen, while a Report is used for formatting, summarizing, and presenting data for printing.
2. Forms are interactive interfaces, whereas Reports are static read-only outputs.

Ans 5. (Dry Run & Output)

A\$	C	X\$	PRINT X\$
"Development"	1	"D"	D
	3	"v"	v
	5	"l"	l
	7	"p"	p
	9	"e"	e
	11	"t"	t
	13	Loop Terminates	

Output: D v l p e t

(Note: PRINT prints each output on a new line. They will be displayed vertically.)

Ans 6. (Debugging QBASIC)

REM to display records	
OPEN "marks.dat" FOR INPUT AS #1	
CLS	
PRINT "Roll Number", "Name", "English", "Nepali"	
DO WHILE NOT EOF(1)	'Correction 1: EOF() should be EOF(1)
INPUT #1, RN, N\$, E, Ne	'Correction 2: NS should be N\$ for string
PRINT RN, N\$, E, Ne	'Correction 3: DISPLAY changed to PRINT
LOOP	'Correction 4: WEND changed to LOOP
CLOSE #1	'Optional but good
practice: CLOSE #1	
END	

Ans 7. (Program Analysis)

(a) The parameters used in the above program are:

Actual Parameters (Arguments): M and N (in the main module).

Formal Parameters: A and B (in the function definition).

(b) The procedure used in the above program is a FUNCTION procedure (specifically, a user-defined function named MUL).

Group 'C'

Ans 8. (Number System)

i) $(00101001)_2 = (?)_{16}$

Group binary into 4s from right to left: 0010 | 1001

Convert to Hexadecimal: 2 | 9

Answer: $(29)_{16}$

ii) $(111)_2 \times (10)_2 - (111)_2$

Multiplication: $111 \times 10 = 1110$

Subtraction: $1110 - 111$

Borrow process: $1110 - 111 = 111$

(Checking in decimal: $7 \times 2 = 14$; $14 - 7 = 7$; 7 in binary is 111)

Answer: $(111)_2$

iii) $(86)_{10} = (?)_2$

Successive division by 2:

$86 / 2 = 43$, Remainder 0

$43 / 2 = 21$, Remainder 1

$21 / 2 = 10$, Remainder 1

$10 / 2 = 5$, Remainder 0

$5 / 2 = 2$, Remainder 1

$2 / 2 = 1$, Remainder 0

$1 / 2 = 0$, Remainder 1

Read remainders from bottom to top.

Answer: $(1010110)_2$

iv) $(1011)_2 \div (10)_2$

$1011 \div 10$:

10 goes into 10: 1 time. (Remainder 0)

Bring down 1: 10 goes into 1: 0 times.

Bring down 1 (makes it 11): 10 goes into 11: 1 time (Remainder 1)

Answer: Quotient = 101, Remainder = 1

Ans 9. (QBASIC Programming)

(a) QBASIC Program (Function & Sub Procedure)

```
DECLARE SUB Average (a, b, c)
DECLARE FUNCTION Sum (a, b, c)

CLS
INPUT "Enter first number: ", num1
INPUT "Enter second number: ", num2
INPUT "Enter third number: ", num3

PRINT "The sum is: "; Sum(num1, num2, num3)
CALL Average(num1, num2, num3)
END

FUNCTION Sum (a, b, c)
    Sum = a + b + c
END FUNCTION

SUB Average (a, b, c)
    avg = (a + b + c) / 3
    PRINT "The average is: "; avg
END SUB
```

(b) QBASIC Program (Sequential Data File)

```
OPEN "record.txt" FOR OUTPUT AS #1
CLS
DO
    INPUT "Enter Roll Number: ", R
    INPUT "Enter Name: ", N$
    INPUT "Enter Address: ", A$
    INPUT "Enter Class: ", C

    WRITE #1, R, N$, A$, C

    INPUT "Do you want to add more records? (Y/N): ", ch$
LOOP WHILE UCASE$(ch$) = "Y"

CLOSE #1
PRINT "Records stored successfully!"
END
```

Ans 10. (C Programming)

Option 1: Check Odd or Even

```
#include <stdio.h>

int main() {
    int num;
    printf("Enter a number: ");
    scanf("%d", &num);

    if(num % 2 == 0) {
        printf("%d is an EVEN number.", num);
    } else {
        printf("%d is an ODD number.", num);
    }
    return 0;
}
```

Option 2: OR (Display series 2, 4, 6... 100)

```
#include <stdio.h>

int main() {
    int i;
    printf("The series is:\n");

    for(i = 2; i <= 100; i = i + 2) {
        printf("%d ", i);
    }
    return 0;
}
```